

AM 463: QUIZ 2
September 19, in class
15 minutes: Closed Book
Answer all problems
Presentation will affect your grade

Question 1:

Estimate the Reynolds number for flow past a chair (take $\nu = 1 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$). Does direction of flow matter? How do you choose the length scale?

Question 2:

Consider steady flow of a very viscous fluid (take $\nu = 0.1 \text{ m}^2 \text{ s}^{-1}$) between two flat plates separated by a space of 5 mm. Assume the pressure gradient is constant and given by $-p_D$ and that the upper plate moves to the LEFT at a speed of 1 cm s^{-1} . Carefully convert to common units and solve the Navier Stokes equations in terms of the free parameter p_D .